EE2005: Electrical Machines
Tutorial 6

1. The full load torque angle of a synchronous motor at rated voltage and frequency is 35° (electrical). Neglect the effects of armature resistance and leakage reactance. If the field current is held constant, how would the full-load torque angle be affected by the following changes in operating condition?

   (a) Frequency reduced by 10 %, load torque and applied voltage constant.

   (b) Frequency reduced by 10 %, load power and applied voltage constant.

   (c) Both frequency and applied voltage reduced by 10%, load torque constant.

   (d) Both frequency and applied voltage reduced by 10%, load power constant.

   [**Ans:** (a) $\delta = 31.07°$ (b) unchanged (c) unchanged (d) $\delta = 39.6°$]

2. A three phase star connected alternator is synchronised with an infinite bus of 11 kV. The prime mover input is then increased such that the output power reaches 15 MW. Now, the field excitation is increased to 130%, when the machine starts operating at 0.8 pf lag. Neglecting the armature resistance, determine:

   (a) synchronous reactance of the machine.

   (b) power factor, load angle and armature current before the field excitation was increased.

   [**Ans:** (a) 2.7406 Ω (b) 0.985 (lead), 19.86°, 800.4 A]

3. A three phase synchronous machine with $X_s = 0.9$ pu on its own rating, is connected via a transmission line to an infinite bus whose voltage is 1 pu. The line reactance is 0.15 pu on the machine base. The internal emf E_f of the machine is kept at 1.35 pu. Calculate

   (a) pull out power of the machine when operating as generator.

   (b) generator terminal voltage and current at pull out.

   (c) reactive powers at the terminals of the generator and at the infinite bus at the moment of pull out.

   [**Ans:** (a) 1.286 pu (b) 0.8786 pu, 1.6 pu (c) -0.568 pu, -0.952 pu]

4. A Y-connected synchronous generator connected to an infinite bus at 12 kV supplies 4000 A at unity power factor. The synchronous reactance is 2 Ω per phase and the stator resistance is negligible. If the prime mover set point is unchanged and the field current is increased by 15 %, determine the new stator current and power factor.

   [**Ans:** 4154 A, 0.9628 (lag)]

5. A 4-pole, 60 Hz, 24 kV, 650 MVA synchronous generator with a synchronous reactance of 1.82 per unit is connected to a grid which can be represented by a 24 kV infinite bus in series with a reactive impedance of j 0.21 Ω. The generator is equipped with a voltage regulator that adjusts the field excitation such that the generator terminal voltage remains at 24 kV independent of the generator loading. The output power of the generator is adjusted to 375 MW.

   (a) Draw a phasor diagram for this operating condition.

   (b) Find the magnitude and phase angle (with respect to the generator terminal voltage) of the generator terminal current.

   (c) Determine the generator terminal power factor.

   (d) Find the magnitude (in p.u and kV) of the generator excitation voltage E_f .

   [**Ans:** (b) 0.578 pu, 3.93° (c) 0.9976 (lag) (d) 1.5 pu, 36 kV]